

The effects of storage temperature and time on the stability of syngas in Polyvinyl fluoride film bags

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ABSTRACT

Polyvinyl fluoride (PVF) film bags are widely used for the temporary storage of air samples prior to analysis due to their durability, toughness, and chemical inertness to a broad range of compounds. However, factors such as temperature and storage time can significantly affect the performance of PVF film bags, with important implications for the accuracy of sample analysis. This study evaluated the stability of syngas in Tedlar® PVF film bags under varying storage temperatures (−20 °C, 25 °C, and 38 °C) and durations (0–168 h), with Pearson correlation analysis conducted to assess the impact of these conditions. The results revealed that both storage temperature and time notably influence syngas stability, particularly for hydrogen and carbon dioxide. A strong negative correlation was observed between the concentrations of these gases and temperature, with the most pronounced negative correlations at 38 °C, showing coefficients of −0.974 and −0.977 for hydrogen and carbon dioxide, respectively. At 25 °C and 38 °C, hydrogen and carbon dioxide exhibited significant concentration losses, with recovery rates of $81.2 \pm 4.5\%$ and $66.9 \pm 1.2\%$ for hydrogen, and $90.9 \pm 1.1\%$ and $89.3 \pm 1.5\%$ for carbon dioxide, respectively. In contrast, gases stored at −20 °C for seven days maintained recovery rates exceeding 95%. This study identifies low-temperature storage in PVF film sampling bags as an effective method for preserving syngas integrity ensuring data accuracy for informed syngas utilization and application decisions.

1. Introduction

Syngas is produced through thermochemical conversion of various feedstocks, including coal, biomass, and waste materials [1,2]. The composition of syngas can vary significantly depending on the chosen feedstock and the specific gasification process employed [3,4]. Syngas primarily consists of carbon monoxide (CO), carbon dioxide (CO₂), hydrogen (H₂), methane (CH₄), and Nitrogen (N₂) [5–8]. Other products are also formed during the gasification process, including char, ash, and oils [9–11]. Syngas can be utilized in various applications within the chemical and energy industries. It is a valuable feedstock for electric gas engines and gas turbines, critical in power generation [12,13]. One notable application of syngas is in methanation, which converts carbon monoxide and hydrogen into synthetic natural gas (SNG). This conversion is essential for the production of SNG, which is used in applications such as ammonia synthesis and fuel cell plants [14–16].

Evaluating syngas composition is crucial for understanding instrument efficiency and process effectiveness. However, one of the challenges in evaluating syngas composition is the limitation of analyzing at

the production site, as it often proves impractical due to the high cost of syngas analysis instruments. Consequently, syngas samples collected on-site in PVF film bags must be transported to the laboratory for analysis. During transport, the samples are exposed to ambient temperatures reaching up to 40 °C within the vehicle [17]. This exposure presents a critical challenge in maintaining the stability of samples.

Furthermore, analysts usually require samples that can be collected within a day. However, given the workload, it may take more time for analysts to examine all samples, necessitating evaluating stability time to estimate the minimum duration required before analysis can commence effectively. Hsieh et al. (2003) suggest a seven-day timeframe is suitable for evaluating the stability of volatile organic compounds (alkenes, alkanes, aromatics, and biogenics). This timeframe encompasses all stages, including sample collection, transportation, and analysis [18]. Improper storage and handling practices can lead to inaccurate data and misinterpretation, potentially impacting the entire process.

The collection of gas samples in canisters is widely practiced [18–20]. These canisters offer several advantages, including inertness,

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ruggedness, and the ability to be reused and maintain durability over time [21]. Despite their high collection efficiency, certain disadvantages become apparent when multiple samples are needed from various sampling points. These include the high costs associated with canisters and the increased difficulty in transporting samples due to their size and weight. In contrast, sampling bags offer a more portable and cost-effective alternative for temporary air sample storage before analysis [22,23]. However, they may suffer from limitations due to the material's weakness and the potential loss of specific components loss during storage [24,25].

Studies, including those by Kim et al. (2012), have shown fluctuations in gas concentrations within PVF film bags. The recovery rates of various compounds vary significantly. For example, methyl isobutyl ketone can retain up to 96% after three days, while styrene and isobutyl alcohol exhibit lower recovery rates of 51% and 31%, respectively [26]. Conversely, PVF film bags effectively retain sulfur compounds, with losses below 5% at 24 h [27]. However, some volatile organic compounds, such as *p*-cresole, indole, and skatole, tend to be lost immediately upon filling the bags [18,28]. Despite these variations, landfill gas samples stored in PVF film bags have shown stability for up to 30 days [29]. Two main factors influence the stability of gas samples stored in PVF film bags: adsorption and diffusion/permeation [28,30,31]. Adsorption refers to gas molecules adhering to the inner surface of the bag material, leading to losses of certain compounds [32]. Conversely, diffusion/permeation describes the movement of gas molecules through the bag material itself [32].

Limited research exists regarding syngas preservation in PVF film bags. Coyne et al. (2011) investigated the stability of carbon dioxide, methane, and carbon monoxide, but only for two days [33]. Other studies reported fluctuation in carbon dioxide and methane concentrations over ten days at temperatures between 20 °C and 30 °C [21]. Evaluation should encompass a wider temperature range and longer storage durations to understand syngas concentration stability better. This study investigated the stability of crucial syngas components such as nitrogen (N₂), hydrogen (H₂), methane (CH₄), carbon dioxide (CO₂), and carbon monoxide (CO) within PVF film sampling bags. The evaluation covered a broader temperature range (−20 °C, 25 °C, and 38 °C) and an extended timeframe (0–168 h). This study aims to elucidate the impact of these storage conditions on syngas recovery rates.

2. Materials and methods

2.1. Materials

The research was conducted using a 3L PVF film sampling bag (Tedlar® bags, CEL Scientific Corp., CA 90703, USA). Standard gas was purchased from the Research Institute of Gas Analytical Science (RIGAS Corp. Korea): hydrogen (8.96 % v/v), carbon monoxide (45.12 % v/v), nitrogen (45.92 % v/v), methane (75.00138 % v/v), and carbon dioxide (24.99862 % v/v). Methane and carbon dioxide were procured from a single container, whereas the remaining gases were sourced from a second, identical container.

2.2. Experimental procedures

Three replicates of syngas-filled sampling bags were stored at different temperatures (−20 °C, 25 °C, and 38 °C) within a controlled environment using a refrigerator and an incubator. The gas concentrations were monitored initially, after 24 h, and then at 12-h intervals up to 168 h. The analysis used a gas chromatograph (iGC 7200, DS Science, Korea) equipped with two columns tailored for specific gas components. A molecular sieve 5 Å column (6 ft x 1/8 inch, 60/80 mesh, inner diameter: 2 mm) was utilized to detect carbon monoxide, hydrogen, and nitrogen. This column was selected for its effectiveness in separating these gases based on their molecular size. A Porapak Q column (6 ft x 1/8 inch, 80/100 mesh, inner diameter: 0.085 inches) was employed for

methane and carbon dioxide analysis, which is well-suited for separating these components. Both columns were interfaced with a thermal conductivity detector (TCD), which was chosen due to its high sensitivity to changes in gas composition. The chromatograph was calibrated before analysis using standard gas of known concentrations, ensuring the quantitative results were accurate. Calibration checks were conducted regularly throughout the experiment to maintain precision and reproducibility, with calibration coefficients of 0.999 for methane and carbon dioxide, 0.991 for hydrogen, 0.997 for carbon monoxide, and 0.998 for nitrogen.

The chromatographic conditions were explicitly optimized for syngas analysis. The carrier gas was helium, with a 25 mL/min flow rate for both columns. The inject and detector temperatures were maintained at 50 °C. The column temperature was maintained at 200 °C, which was crucial for the accurate resolution of each gas component within a total run time of 20 min. These conditions allowed for the effective separation and quantification of all identified compounds, providing reliable data on the stability of the syngas components over time and under varying storage conditions.

2.3. Statistical analysis

The correlation between gas concentration and storage time was analyzed using Pearson correlation (95% confidence level) at each temperature using R software. In order to support the correlation, the differences between gas concentrations among three different temperatures were also analyzed at each interval of time. The significant differences in storage time and temperature were statistically analyzed using one-way ANOVA (95% confidence level). Furthermore, the differences in each group are continued using the Tukey test (95% confidence level).

3. Results

3.1. The effects of storage temperature and period on syngas concentration

Fig. 1 shows the dynamics of syngas concentration at different temperatures over 168 h. We also investigated the concentration differences between all temperatures at each time interval. Nitrogen, carbon monoxide, and methane showed fluctuating concentrations but tended to be constant over 168 h (Table S1 and S.2). For nitrogen, the concentrations at 132 h in the −20 and 38 °C storage conditions and 156 h in the 25 °C storage condition was significantly different from the initial concentration ($p < 0.05$), but there was no significant difference between the final concentration and the initial concentration at 168 h ($p > 0.05$). In the case of methane, the concentration at 144 h in −20 °C was significantly different from the initial concentration ($p < 0.05$), but there was no significant difference between the final concentration and the initial concentration at 168 h ($p > 0.05$). Carbon monoxide showed stable concentration, and no significant differences were observed during 168 h at −20, 25, and 38 °C storage conditions. Overall, an insignificant difference at the end of time storage (168 h) indicates the stability of syngas concentration over time ($p > 0.05$). Moreover, there were no significant correlations between carbon monoxide concentration and storage time at all evaluated temperatures. Nitrogen and methane also showed no significant correlation between concentration and storage time at −20 and 38 °C ($p > 0.05$). Fluctuating concentrations of each gas may cause these significant differences at specific times in the same PVF film bags. The observed fluctuation in gas concentrations is likely attributed to the influence of the unstable gas among the mixed gases, as carbon dioxide was mixed with methane, and nitrogen was mixed with hydrogen and carbon monoxide. Since the concentration of the unstable gas can easily change over time, it can also lead to a coupled change in the concentration of the relatively stable gas.

Hydrogen and carbon dioxide showed different results. Hydrogen

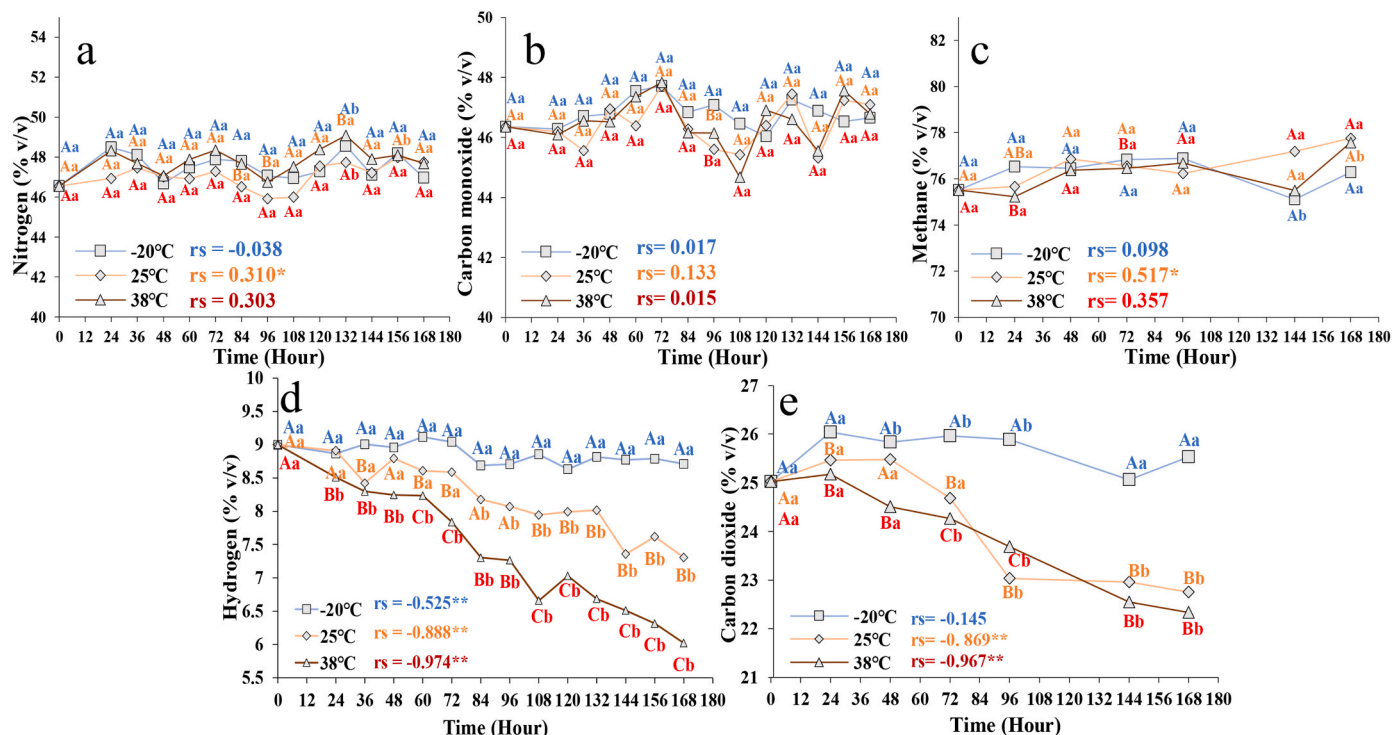


Fig. 1. Syngas concentration at different temperatures over 168 h: (a) Nitrogen; (b) Carbon monoxide; (c) Methane; (d) Hydrogen; (e) Carbon dioxide. Different lowercase letters in the same color are significantly different from initial concentration at each temperature. Uppercase shows the difference of concentration between storage temperatures at the same storage time.

rs = Correlation coefficient

* = Significantly different ($p < 0.05$); ** = Significantly different ($p < 0.01$). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

showed a significant negative correlation between concentration and storage time over 168 h at all three evaluated storage temperatures, while carbon dioxide exhibited a significant negative correlation between concentration and storage time at 25 °C and 38 °C, but not at -20 °C ($p < 0.01$). Increasing temperature linearly decreases gas concentration. Pearson correlation analysis results clearly show an increased linear correlation coefficient with increasing temperature. The high decreasing concentration starting from the temperature of 25 °C, hydrogen, and carbon dioxide have a coefficient correlation of -0.888 and -0.869, respectively. Furthermore, the temperature of 38 °C shows the concentration of gases dropped intensely for 168 h, with a correlation coefficient of -0.974 for hydrogen and -0.967 for carbon dioxide. At 38 °C storage condition, hydrogen began to show a significant difference from its initial concentration after 24 h, while carbon dioxide began to show a significant difference from its initial concentration after 72 h. It indicates temperatures affect concentration.

Table S3 shows the sum of gas concentrations in the same PVF film bags. It keeps approximately a 100% v/v because the concentrations in certain gases decrease following the increase in other gases in the same PVF film bags. This indicates a balance of gases within the syngas. Fig. 1 displays a decrease in hydrogen concentration, accompanied by a subsequent increase in nitrogen and carbon monoxide concentrations throughout the time intervals. These trends are proven by the correlation coefficients between the concentration of hydrogen and nitrogen (table S4). At 25 and 38 °C, the correlation coefficients were -0.307 and -0.281, respectively, signaling a shift in concentration in the gas constituents within the PVF film bags. Likewise, there is a marked reduction in carbon dioxide concentration, accompanied by an increase of methane concentration, particularly at 25 and 38 °C, as denoted by their correlation coefficients (-0.401 and -0.264) (Table S5). These changes underscore the losses of gases in the PVF film bags, influencing the gas composition over time. The loss of hydrogen and carbon dioxide during

storage conditions shifted the concentration to other gasses in the same PVF film bags.

3.2. Recovery rate after seven days

The stability of syngas (recovery rate) over seven days is shown in Fig. 2. Nitrogen, carbon monoxide, and methane have shown a high percentage recovery. Even though there were fluctuations in each interval, the percentage of recovery remained consistent. Nitrogen, carbon monoxide, and methane have recovery rates above 98%.

The other pattern was observed with hydrogen and carbon dioxide, which exhibited unstable concentrations (Table S6 and S.7). For hydrogen, storage at -20 °C showed stable recovery rates of 98.5–100.5% up to the 3rd day, but this decreased to 96.8 ± 0.8% on the 4th day. When stored at 25 °C, the recovery rates remained above 95% for the first three days but decreased to 89.7% ± 4.9% on the 4th day. At 38 °C, the recovery rates decreased sharply on the 2nd day to 91.6 ± 1.5%. The high carbon dioxide recovery rates (approximately 100%) were maintained until the 7th day at -20 °C. At 25 °C, a stable recovery rate was observed until the 3rd day, after which it decreased sharply to 92.0 ± 1.1%, 91.8 ± 2.9%, and 90.9 ± 1.1% at day of the 4th, 6th, and 7th days, respectively. The effect of temperature clearly showed at 38 °C, the recovery rate on the 1st and 2nd day tended to be high (100.6 ± 0.2% and 97.9 ± 1.5%) but then decreased linearly with increasing storage time. Overall, hydrogen had a final percentage recovery of 81.2 ± 4.5% and 66.9 ± 1.2%, while carbon dioxide had 90.9 ± 1.1% and 89.3 ± 1.5% at 25 °C and 38 °C, respectively. Both gases recovered highly in temperatures -20 °C, despite fluctuations that tend to be constant at over 95%. Hydrogen has a recovery rate of 96.8 ± 1.3%, and carbon dioxide 102 ± 0.2% after seven days.

Hydrogen and carbon dioxide storage in sampling bags must consider the effects of temperature and time. In this study, -20 °C

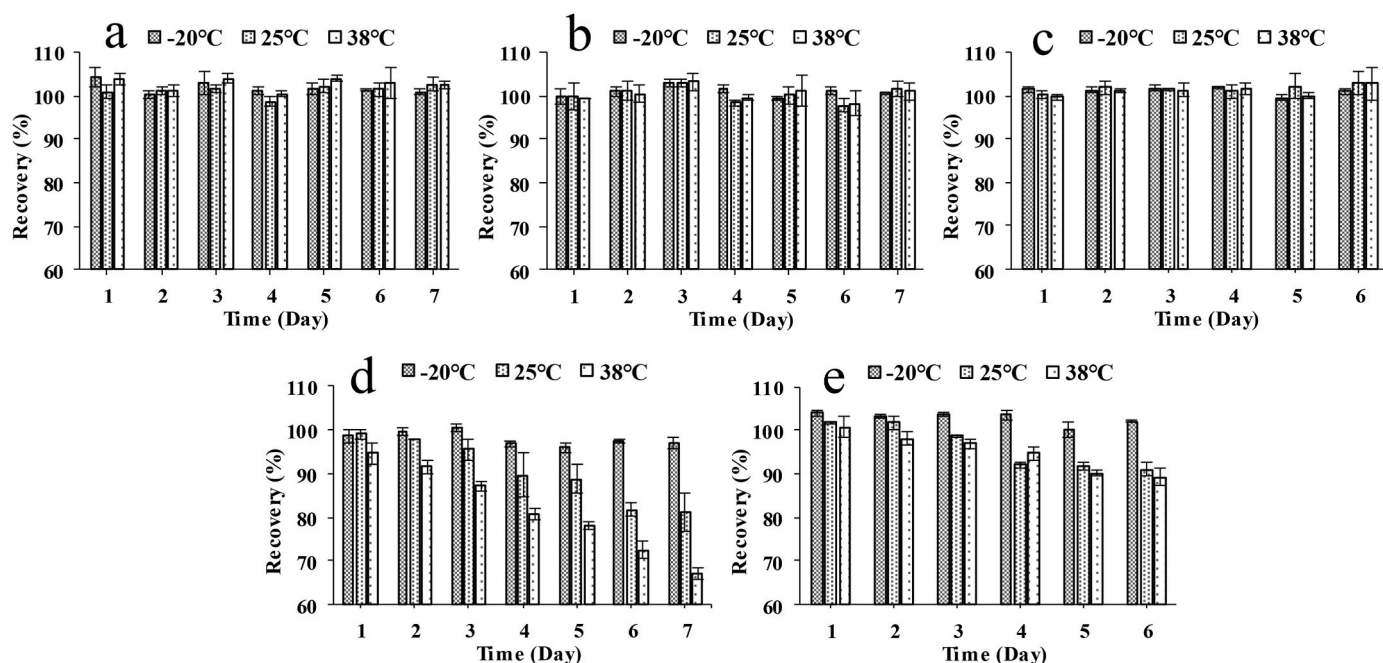


Fig. 2. Syngas recovery (%) at different temperatures: (a) Nitrogen; (b) Carbon monoxide; (c) Methane; (d) Hydrogen; (e) Carbon dioxide.

appears to have the most significant recovery in hydrogen and carbon dioxide percentages. Despite limited studies about the reliability of syngas in PVF sampling bags, the stability of carbon monoxide, carbon dioxide, and methane concentrations was reported by Coyne et al. (2011) with a recovery of 90%, 100%, and 99%, respectively, after two days storage on PVF sampling bags [33]. In another study by Silva et al. (2016), the fluctuation of methane recovery (97.3–102.6%) after ten days occurs when sampling in ambient temperature at different areas [21]. Similar results reported the high recovery of some odorants when stored at -20°C [34]. The compounds in sampling bags were more stable and reproducible at low temperatures, probably because the reaction rate was lower [35].

4. Discussion

PVF film bags are commonly used for gas storage before analysis. However, the study on syngas stability in PVF film bags was limited. Coyne et al. (2011) reported a 100% recovery of carbon dioxide after two days, which is in line with our findings of 103.2 ± 0.4 , 101.8 ± 0.9 , and $97.9 \pm 1.5\%$ recovery at temperatures of -20°C , 25°C , and 38°C , respectively, over the same duration [36]. However, our study conducted extended observation and revealed a notable decrease in recovery over seven days, particularly at 25°C and 38°C , with recoveries dropping to $90.9 \pm 1.1\%$ and $89.3 \pm 1.5\%$, respectively. Another study by Silva et al. (2015), which involved sampling atmospheric gases, found a high recovery rate of carbon dioxide (99–100%) after ten days of storage at 20 – 30°C [37]. Ajhar et al. (2010) found that the permeation of carbon dioxide influenced the stability of this gas in PVF film bags [29]. In the other gasses, methane and carbon monoxide showed recovery rates comparable to previous research [36,37], with a high recovery of these gases (above 97%).

The stability of gases within storage bags is contingent upon a variety of factors, such as bag material characteristics, including film thickness and permeation coefficient, as emphasized by Ghimenti et al. (2015) and Mochalski et al. (2013) [25,31]. Our research contributes to the existing knowledge by demonstrating that low temperatures substantially improve the stability of syngas by slowing down permeation and reducing physical adsorption. This aligns with reported reductions in the loss of acetic and phenolic compounds at lower temperatures (-20°C),

as identified in previous research [23]. In another study, black PVF film bags achieved a high carbon dioxide recovery rate (approximately 100%) over 10 days [37]. Therefore, further study is needed to investigate the effect of light exposure on the stability of syngas. The harmony of our study with prior research, coupled with new insights into syngas stability, underscores the intricate balance between physical adsorption and permeation, including diffusion in the context of gas storage.

The implications of this study are far-reaching, both scientifically and practically. Scientifically, it contributes to a deeper understanding of gas storage dynamics, particularly for complex gas mixtures like syngas. The findings provide a framework for predicting and controlling the stability of syngas in various storage conditions, which is paramount for accurate analytical results. This research offers practical solutions for industries and laboratories that frequently utilize or analyze syngas. By identifying optimal storage conditions (low temperatures and suitable storage durations), this study guides practitioners in maintaining the integrity of syngas samples, thereby ensuring the reliability and accuracy of subsequent analyses. Enhancing syngas stability in PVF film bags under low-temperature conditions is crucial for various applications, including environmental monitoring, industrial processing, and research laboratories, where precise and reliable gas analysis is essential.

5. Conclusions

In this work, the authors evaluated the long-term storage of syngas (7 days) at different temperatures in PVF film bags. We found that temperatures play significant roles in syngas concentration, mainly hydrogen and carbon dioxide. The percentage recovery of hydrogen decreased linearly with the storage time at both 25°C and 38°C . This study suggests storing hydrogen for three days at -20°C and only a maximum of 1 day at 25°C . In contrast, carbon dioxide can be stored for up to 7 days at two storage temperatures (-20°C , 25°C) without significant recovery loss. However, storage at 38°C for more than one day is not recommended due to a noticeable drop in recovery. In the case of nitrogen, carbon monoxide, and methane, this study found a stable concentration over seven days. However, since the high percentage recovery, we recommended storing these gases at -20°C and 25°C . These findings suggest that careful consideration of storage temperatures is

crucial for maintaining the integrity of syngas components before analysis. Future research could explore the other effects such as other storage materials to optimize syngas preservation further.

CRediT authorship contribution statement

Riuh Wardhani: Writing – original draft, Visualization, Validation, Software, Data curation. **Woohee Chung:** Investigation, Formal analysis. **Jinho Shin:** Validation, Software, Investigation. **Heekwon Ahn:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Chat GPT to improve language and readability to deal with high academic writing standards. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the publication's content.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2024.08.452>.

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